Dr. Mohamed Aziz BOUKRAA – Curriculum Vitae

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Research Interests

Mathematical Modeling, Inverse Problems, Wave Propagation, Seismic Imaging, Full Waveform Inversion, Numerical Analysis, Partial Differential Equations, Finite Element Method, Scientific Computing, High Performance Computing, Applications to realistic environments.

Education

2018–2021 Doctorate in Applied Mathematics, Laboratory of Mathematics Nicolas Oresme (LMNO), University of Caen Normandy, France.
Title: *Méthodes inverses à régularisation évanescente pour l'identification de conditions aux limites en théorie des plaques minces.*
English title: *Fading regularization inverse methods for the identification of boundary conditions in thin plate theory.*
Thesis supervisor: Pr. Franck Delvare. **Defense:** December 14, 2021.
2016–2018 Master's degree in Mathematics, Spec. Optimization and Mathematical Physics, University of Toulon, France.
2013–2016 Bachelor's degree in Mathematics, University Jean Monnet, Saint-Étienne, France.

Experience

Research Experience

Sep 2024 – Present Postdoctoral Research Fellow, Department of Informatics, University of Oslo, Norway. *Research in inverse problems and image reconstruction in acoustics and elastography.* This position is part of the MSCA CO-FUND DSTrain programme. The project is carried out in collaboration with the Digital Signal Processing group (DSB), IFI, University of Oslo.
Sep 2022 – Aug 2024 Postdoctoral position. INRIA (IDEFIX Team), Ensta Paris, Institut Polytechnique de Paris, and EDF R&D. *Development of inverse problem methodologies for interface imaging, application to the rock-concrete interface for a hydroelectric dam.*

Sep 2021 – Aug 2022 Contractual assistant professor ("ATER"). Laboratory of Mathematics Blaise Pascal, Polytech Clermont INP.

Sep 2018 – Dec 2021 Doctoral thesis. Laboratory of Mathematics Nicolas Oresme (LMNO), University of Caen Normandy, France. PDF link.

Mar 2018 – Aug 2018 Research internship (MSc). Alten SA, Alten Innovation Center, Chaville, France.

Mar 2017 – Jun 2017 Research internship (MSc). LIS Laboratory, University of Toulon, France.

Teaching Experience

Sep 2025 – 2027 Department of Informatics, University of Oslo, Norway. Contractual assistant professor.

- Lectures in "Ultrasound Imaging" course: Advanced Topics module.
- Lectures in "Statistical Signal Processing" course: Random processes and signal modeling modules.

Sep 2021 – Aug 2022 Polytech Clermont INP. Contractual assistant professor (192h).

- Tutorials in Mathematics, Probability, Statistics, and Numerical Analysis.
- Lectures and tutorials for Architect-Engineer double degree.

Oct 2019 – Sep 2021 University of Caen Normandy. Part-time teacher (75h).

- Lectures and tutorials in Analysis, Complex Numbers, Geometry.

Skills

Languages: French and Arabic (Native), English (Fluent), Norwegian (B1 - Intermediate), Spanish (Basic).
Software: Matlab, FreeFEM, PETSc, MPI, Python, Maple, LaTeX, Gmsh.

International Collaborations

- **Aurora Grant (Research Council of Norway) 2026– 2027:** FWI-Dams: Full-Waveform Inversion for Dam Safety. Mobility grant to establish and expand contact between French and Norwegian research organizations.
- **PHC Utique Project, 2020 & 2022:** Visiting Researcher at the National School of Engineers of Tunis. Collaboration with LAMSIN laboratory on finite element algorithms with fading regularization for inverse problems.

International Schools

- Bath Symposium on Inverse Problems and AI in Medicine, University of Bath, UK, June 2025. `https://bathsymposium.ac.uk/symposium/inverse-problems-and-artificial-intelligence-in-medicine/`
- Geilo Winter School 2025 on Inverse Problems, SINTEF/NTNU, Geilo, Norway, Jan 2025. `https://www.sintef.no/projectweb/geilowinterschool/2025-inverse-problems/`
- French–Romanian Summer School on Applied Mathematics, University of Bucharest, Romania, July 2019. `https://sites.google.com/site/marinliviu/french-romanian-summer-school-on-applied-mathematics/6th-french-romanian-summer-school-on-applied-mathematics-3-11-july-2019`

Publications and Communications

Journal Papers

1. **Boukraa, M. A.**, Karabiyik, Y., Austeng, A., Holm, S., Näsholm, S. P.. Proof of concept for full-waveform inversion in ultrasound time-harmonic shear-wave elastography. *Physics in Medicine & Biology, 2026.*: 10.1088/1361-6560/ae6af4
2. **Boukraa, M. A.**, Delvare, F., Caillé, L.. Fading regularization method for an inverse boundary value problem associated with the biharmonic equation. *Journal of Computational and Applied Mathematics, 440, 115755, 2025.*: 10.1016/j.cam.2024.116285
3. **Boukraa, M. A.**, Amdouni, S., Delvare, F.. Fading regularization FEM algorithms for the Cauchy problem associated with the two-dimensional biharmonic equation. *Mathematical Methods in the Applied Sciences, 46(2), 2389–2412, 2023.*: 10.1002/mma.8651

Conference Proceedings

1. **Boukraa, M. A.**, Audibert, L., Bonazzoli, M., Haddar, H., Vautrin, D.. High-Resolution Seismic Imaging for Dam-Rock Interface using Full-Waveform Inversion. *Waves 2024, Berlin, Germany, 2024.*: `10.17617/3.MBE4AA`
2. **Boukraa, M. A.**, Bonazzoli, M., Haddar, H., Audibert, L., Vautrin, D.. Imaging a dam-rock interface with inversion of a full elastic-acoustic model. *11th International Conference on Inverse Problems in Engineering (ICIPE 2024), Rio de Janeiro, Brazil, 2024.*: `https://uca.hal.science/hal-04661884`

3. **Boukraa, M. A.**, Audibert, L., Bonazzoli, M., Haddar, H., Vautrin, D.. Imagerie dinterface barrage-fondation par inversion de forme donde complète. *E3S Web of Conferences, 2024.*: 10.1051/e3sconf/202450404002

Talks in International Conferences

- 49th Scandinavian Symposium on Physical Acoustics (SSPA 2026), Geilo, Norway, January 2026. https://www.norskfysisk.no/faggrupper/faggruppe-akustikk/sspa/
- 11th International Conference on Inverse Problems, Control and Shape Optimization (PICOF 2025), Hammamet, Tunisia, October 2025. https://picof2025.sciencesconf.org/
- IEEE International Ultrasonics Symposium (IUS 2025), Utrecht, Netherlands, April 2025. https://2025.ieee-ius.org/
- Ultrasonic Imaging and Tissue Characterization Conference (UITC 2025), Silver Spring, MD, USA, April 2025.
- 48th Scandinavian Symposium on Physical Acoustics (SSPA 2025), Geilo, Norway, January 2025. https://www.norskfysisk.no/faggrupper/faggruppe-akustikk/sspa/
- WAVES 2024 (16th International Conference on Mathematical and Numerical Aspects of Wave Propagation), Berlin, Germany, June 2024. https://waves2024.mps.mpg.de/
- 11th International Conference on Inverse Problems in Engineering (ICIPE24), Rio de Janeiro, Brazil, June 2024. https://icipe2024.org/
- 11th International Conference on Inverse Problems in Modeling and Simulations (IPMS 2024), Malta, May 2024. https://www.ipms-conference.org/ipms2024/
- 11th Applied Inverse Problems Conference (AIP 2023), Göttingen, Germany, September 2023. https://aip2023.de/
- 10th International Conference on Inverse Problems, Control and Shape Optimization (PICOF'22), Caen, France, October 2022. https://picof22.sciencesconf.org/
- 10th International Conference on Inverse Problems in Engineering (ICIPE20), Francavilla al Mare (Chieti), Italy, May 2022. https://iopscience.iop.org/article/10.1088/1742-6596/2444/1/011001

Other Talks

- Improved full-waveform inversion strategies for ultrasound time-harmonic elastography, Oslo, Norway, dScience Lunch Seminar, March 2026. https://www.uio.no/dscience/english/news-and-events/events/26-ls-mar5.html
- Journées Scientifiques AGAP Qualité Géophysique Appliquée, Poitiers, France, March 2024. https://www.agapqualite.org/2023/07/11/journees-scientifiques-agap-26-au-28-mars-2024/
- Congrès des Jeunes Chercheurs en Mécanique (Méca-J 2023), Online, August 2023. https://mecaj2023.sciencesconf.org/
- 11ème Biennale de la Société de Mathématiques Appliquées et Industrielles (SMAI 2023), Le Gosier, Guadeloupe, France, May 2023. https://smai2023.math.cnrs.fr/fr/
- Congrès Français de Mécanique (CFM 2022), Nantes, France, August 2022. https://cfm2022.fr/
- 45ème Congrès National d'Analyse Numérique (CANUM 2022), Évian-les-Bains, France, June 2022. https://canum2020.math.cnrs.fr
- Congrès des Jeunes Chercheurs en Mécanique (Méca-J 2021), Online, August 2021. http://meca-j.sciencesconf.org
- 10ème Biennale de la Société de Mathématiques Appliquées et Industrielles (SMAI 2021), La Grande-Motte, France, June 2021. https://smai2021.math.univ-toulouse.fr
- 9ème Biennale de la Société de Mathématiques Appliquées et Industrielles (SMAI 2019), Guidel-Plages (Morbihan), France, May 2019. http://smai.emath.fr/smai2019/index.php

Activities of Collective Interest

- Participation at *"Semaines d'Etudes Mathématiques et Entreprises" (SEME)*, Pointe-à-Pitre, Guadeloupe, May 15–19, 2023. semeantilles.sciencesconf.org Joint Work: "Optimal distribution for steam production". Industry Collaboration: Refinery Industry SARA (Pôle 972, Martinique).
- Organizing committee member of the international conference **PICOF22**, Caen, October 25–27, 2022. picof22.sciencesconf.org
- Organization of the workshop *"Chercheurs-Chercheuses"*, Dôme de Caen, February 2, 2020.